

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12405570

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Bayan; Heshig et al.

Method and device for compensating surface error of holographic grating substrate

Abstract

A method and a device for compensating a surface error of a holographic grating substrate based on scanning and exposure technology relates to a technical field of grating development. The method and the device adopt surface error compensation technology based on phase modulation of interference fringes of a scanning beam interference lithography system. The method and the device solve a poor grating diffraction wavefront quality caused by a surface processing error in a field of holographic grating research and improves full-aperture diffraction wavefront quality of a grating. Through coordination between feedback of the displacement measurement apparatus for the position of the two-dimensional worktop and the modulation of the scanning beam interference lithography system, full-aperture compensation of the surface error of the holographic grating substrate is achieved, which not only reduces processing accuracy requirements for the holographic grating substrate, but also improves the full-aperture diffraction wavefront quality of the grating.

Inventors: Bayan; Heshig (Changchun, CN), Jiang; Shan (Changchun, CN), Li; Yubo (Changchun, CN), Liu; Zhaowu (Changchun, CN), Wang; Wei (Changchun, CN), Li; Wenhao (Changchun, CN), Li; Shuo (Changchun, CN), Jiang; Yanxiu (Changchun, CN)

Applicant: Changchun Institute Of Optics, Fine Mechanics And Physics, Chinese Academy Of Sciences (Changchun, CN)

Family ID: 87953785

Assignee: CHANGCHUN INSTITUTE OF OPTICS, FINE MECHANICS AND PHYSICS, CHINESE ACADEMY OF SCIENCES (Changchun, CN)

Appl. No.: 18/812069

Filed: August 22, 2024

Prior Publication Data

Document Identifier

Publication Date

Foreign Application Priority Data

CN202311055791.3

Aug. 22, 2023

Publication Classification

Int. Cl.: G03H1/04 (20060101); G03H1/00 (20060101)

U.S. Cl.:

CPCG03H1/0486 (20130101); G03H1/0005 (20130101); G03H2001/0094 (20130101)

Field of Classification Search

CPC: G02B (6/02138); G02B (27/0172); G02B (5/1847); G02B (5/1857); G02B (27/0025); G03F (7/70775); G03F (7/70408); G03H (1/202); G03H (1/00); G03H (1/0005); G03H (1/0248); G03H (1/0276); G03H (1/0408); G03H (1/0465); G03H (1/0486); G03H (1/20); G03H (1/2294); G03H (1/26); G03H (2001/026); G03H (2001/0439); G03H (2001/0482); G03H (2001/205); G03H (2001/2226); G03H (2222/36); G03H (2223/16); G03H (2223/24); G03H (2225/21); G03H (2225/32); G03H (2260/00); G03H (2001/0094); G03H (2001/0491)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6882477	12/2004	Schattenburg	359/35	G03F 7/70408
2004/0005116	12/2003	Sweetser	385/37	G02B 6/124
2009/0027640	12/2008	Shibazaki	356/450	G03F 7/70775
2009/0257039	12/2008	Kishikawa	355/71	G03B 27/72
2015/0378306	12/2014	Gurevich	359/12	G03H 1/0005
2018/0284619	12/2017	Shirato	N/A	G03F 7/2057
2022/0228890	12/2021	Li	N/A	G01B 11/02
2024/0061352	12/2023	Wang	N/A	G03F 7/2006

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104237987	12/2013	CN	N/A
104570621	12/2014	CN	N/A
105954977	12/2015	CN	G02B 5/1857
107607060	12/2017	CN	N/A
112255717	12/2020	CN	G02B 5/1852
112255722	12/2020	CN	N/A

OTHER PUBLICATIONS

Machine translation of CN-105954977-A (Year: 2016). cited by examiner

Machine translation of CN-112255717-A (Year: 2021). cited by examiner

Notification to Grant Patent Right for Invention, Chinese Application No. 202311055791.3, mailed Oct. 23, 2023 (3 pages). cited by applicant

CNIPA, Office Action issued for Chinese Application No. 202311055791.3, mailed Sep. 19, 2023 (12 pages). cited by applicant

Chao Yang et al., "Reducing machine-ruling grating wavefront value by compensating grating substrate surface error in real time", «Optics Communications», full text, Issued Date Mar. 8, 2018. cited by applicant

Zhaowu Liu et al., "Study on wavefront control of grating diffraction on Scanning Beam Interference Lithography System", Chinese doctoral dissertation full text database Basic Science series, Text pages 83-105, Issued Date Aug. 15, 2017. cited by applicant

Chao Yang et al., "Research on correcting of the grating groove error in large area and real-time monitoring technology", Chinese doctoral dissertation full text database Engineering Science and Technology II series, Text pages 83-105, Issued Date Oct. 15, 2015. cited by applicant

Primary Examiner: Sun; Pinping

Assistant Examiner: Lee; Matthew Y

Attorney, Agent or Firm: HOWARD M COHN and Associates, LLC

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a technical field of grating development, and in particular to a method and a device for compensating a surface error of a holographic grating substrate based on scanning and exposure technology.

BACKGROUND

(2) Diffraction wavefront quality is one of the important performance indicators of gratings. High diffraction wavefront quality enables the gratings to have extremely high resolution and imaging characteristics. Therefore, the gratings are widely used in high-precision equipment such as astronomical telescopes, spectrometers, and high-energy laser systems. Grating diffraction wavefront errors mainly consist of two parts, which are substrate surface error and ruled line errors. Difficulty of processing grating substrates increases with sizes of the grating substrates. In addition, regardless of the sizes of the grating substrates, it is impossible to achieve near-ideal processing accuracy, resulting in the substrate surface error. For small-sized gratings, it is easy to achieve high surface accuracy, but for large-sized grating substrates, when requirements for the diffraction wavefront quality are extremely high, there are significant challenges such as high processing difficulty, long duration, high cost, and limited surface accuracy in a surface of the large-sized grating substrates surface, making it difficult to ensure extremely low surface error over meter-scale dimensions. Therefore, in order to obtain a near-ideal diffraction wavefront, under a condition of known substrate surface error, the diffraction wavefront is modulated by controlling the ruled line errors to achieve an effect of compensating for the surface error of the grating substrates.

(3) There are two technical solutions in the prior art. In a first technical solution thereof, CHAO YANG proposed a method for compensating surface error of a grating substrate based on a

mechanical engraving machine by using a substrate support and by grooving on the grating substrate. First, optimal positions of four support seats under a loading base of the grating substrate are optimized through simulation of a finite element software to minimize a surface change of the loading base of the grating substrate. Then, a grating groove active control technology is applied to compensate for the surface error of the grating substrate by changing a swing angle between a diamond engraving knife and a surface of the grating substrate. CHAO YANG conducted a grating diffraction wavefront compensation experiment, and it turns out that a peak-valley (PV) value of a +36th order diffraction wavefront of a medium-step grating dropped from 0.076λ to 0.056λ , verifying the feasibility of compensating for the surface error in the engraved grating. A second technical solution thereof is to reduce the surface error in active control of the grating diffraction wavefront. ZHAOWU LIU proposed an active wavefront control technology based on a scanning beam interference lithography system. A sinusoidal target wavefront is simulated and predetermined. By controlling a phase distribution in an X direction and conducting exposure and development experiments, a grating mask with a sinusoidal wavefront consistent with the sinusoidal target wavefront is obtained.

(4) However, the prior art has some defects. First, in a field of ruled grating research and development, online correction of a curvature of a ruled line is limited by a working mode of a grating ruling machine itself. A compensation amount of the surface error obtained by correcting the curvature of a single ruled line is extremely small, which is unable to compensate for larger surface error and does not belong to a technical field of holographic grating research and development. Second, in terms of wavefront modulation, the prior art only achieves phase modulation in the X direction (i.e., a stepping direction) and does not achieve the phase modulation in a Y direction (i.e., a scanning direction). The prior art fundamentally does not have a full-aperture compensation function for the surface errors of a holographic grating substrate.

SUMMARY

(5) To solve problems in the prior art, the present disclosure provides a method and a device for compensating a surface error of a holographic grating substrate based on scanning and exposure technology.

(6) In a first aspect, the present disclosure provides the method for compensating the surface error of the holographic grating substrate based on the scanning and exposure technology. The method comprises steps: obtaining a surface error distribution diagram corresponding to the holographic grating substrate by detecting the surface error of the holographic grating substrate; obtaining a phase distribution of predetermined phases compensating for the surface error through simulation calculation; and decomposing the phase distribution of the predetermined phases into a first coordinate matrix corresponding to phases of the holographic grating substrate in a point-to-point manner, exposing coordinate points of the predetermined phases, different with the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on a surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain a grating mask.

(7) In one optional embodiment, after the step of obtaining the phase distribution of the predetermined phases compensating for the surface error through simulation calculation, the method further comprises a step of pre-compensating phases of the surface error by using a scanning interference field exposure model, and forming a compensation table by pre-compensated phase values of the phases of the surface error.

(8) In one optional embodiment, the step of decomposing the phase distribution of the predetermined phases into the first coordinate matrix corresponding to phases of the holographic grating substrate in the point-to-point manner, exposing the coordinate points of the predetermined phases, different with the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on the surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain the grating mask comprises: accurately positioning the

holographic grating substrate of a predetermined size from an exposure start point to an exposure end point by a displacement measurement apparatus to obtain a second coordinate matrix of the holographic grating substrate; and corresponding the predetermined phases for compensating the surface error of the holographic grating substrate one-to-one to the coordinate points in the second coordinate matrix of the holographic grating substrate, feeding back a position of the holographic grating substrate by the displacement measurement apparatus in real time, and scanning and exposing the photoresist on the surface of the holographic grating substrate by a scanning beam interference lithography system, and then developing the photoresist to obtain the grating mask after surface compensation modulation.

(9) In one optional embodiment, a phase modulation principle of the scanning beam interference lithography system is as follow.

(10) First, spatially filtered and collimated small-size Gaussian beams emitted by the scanning beam interference lithography system are interfered at beam waist positions thereof to form interference fringes. A phase of the interference fringes at a position (x,y) at time t is $\phi_{\text{sub.ph}}(x,y,t)$, a total exposure after scanning and exposure is:

$$(11) D(x,y) = A(x,y) \{ 1 + \gamma \cos[\frac{2}{p}x + \phi_{\text{ph}}(x,y)] \} \cdot \frac{2}{p} + \phi_{\text{ph}}(x,y)$$

is an exposure phase distribution after stepping, scanning, and exposure, and $\Phi_{\text{sub.ph}}(x,y)$ is a predetermined phase distribution related to $\phi_{\text{sub.ph}}(x,y,t)$, and a value of $\phi_{\text{sub.ph}}(x,y,t)$ is changed to obtain the predetermined phase distribution $\Phi_{\text{sub.ph}}(x,y)$ of the surface of the holographic grating substrate.

(12) Then, detecting the surface error of the holographic grating substrate, and obtaining the phase distribution of the predetermined phases compensating for the surface error through simulation calculation:

$$(13) \phi_{\text{ph,e}}(x,y) = \begin{bmatrix} \phi_{\text{ph,e}}(x_1,y_1) & \dots & \phi_{\text{ph,e}}(x_m,y_1) \\ \vdots & \ddots & \vdots \\ \phi_{\text{ph,e}}(x_1,y_n) & \dots & \phi_{\text{ph,e}}(x_m,y_n) \end{bmatrix}.$$

(14) Measuring a displacement of a two-dimensional worktop by a displacement measurement apparatus and obtaining the second coordinate matrix H of the holographic grating substrate:

$$(15) H = \begin{bmatrix} x_1, y_1 & \dots & x_m, y_1 \\ \vdots & \ddots & \vdots \\ x_1, y_n & \dots & x_m, y_n \end{bmatrix}.$$

(16) During a scanning and stepping process of the two-dimensional worktop, when not modulated, there is a phase distribution W of the holographic grating substrate one-to-one corresponding to coordinate points in the second coordinate matrix H; the phase distribution W is represented by:

$$(17) W = \begin{bmatrix} \phi_{\text{sub.ph}}(x_1,y_1) & \dots & \phi_{\text{sub.ph}}(x_m,y_1) \\ \vdots & \ddots & \vdots \\ \phi_{\text{sub.ph}}(x_1,y_n) & \dots & \phi_{\text{sub.ph}}(x_m,y_n) \end{bmatrix}.$$

(18) ϕ is a phase modulation term and is a feedback of a phase control apparatus through a displacement measurement of the two-dimensional worktop. Phases of the interference fringes are directionally modulated by changing frequencies of phase modulators, ϕ is expressed as:

$$(19) \phi = \begin{bmatrix} \phi_{(1,1)} & \dots & \phi_{(m,1)} \\ \vdots & \ddots & \vdots \\ \phi_{(1,n)} & \dots & \phi_{(m,n)} \end{bmatrix}.$$

(20) After phase modulation, a phase distribution W.sub.n of modulated phases one-to-one corresponding to the coordinate points in the second coordinate matrix is represented by:

$$(21) W_n = \left[\begin{array}{cc} \begin{array}{c} (x_1, y_1) \\ + \\ (1, 1) \end{array} & \begin{array}{c} \text{.Math.} \\ (x_m, y_1) \\ + \\ (m, 1) \end{array} \\ \begin{array}{c} \text{.Math.} \\ (x_1, y_n) \\ + \\ (1, n) \end{array} & \begin{array}{c} \text{.Math.} \\ (x_m, y_n) \\ + \\ (m, n) \end{array} \end{array} \right] = \text{ph}, e(x, y).$$

(22) The phase distribution $W_{\text{sub},n}$ of the modulated phases is same as the phase distribution $\Phi_{\text{sub},\text{ph},e}(x,y)$ of the predetermined phases.

(23) In a second aspect, the present disclosure provides the device for compensating the surface error of the holographic grating substrate based on the scanning and exposure technology. The device has scan and exposure functions and is applied to the method mentioned above. The device comprises a first reflector, a second reflector, a third reflector, a fourth reflector, a first phase modulator, a second phase modulator, a first beam splitter, a second beam splitter, a third beam splitter, a two-dimensional worktop, a reference mirror, a displacement detection apparatus, a phase detector, and a master controller.

(24) A left exposure beam is reflected by the first reflector and the fourth reflector in sequence, passes through the first phase modulator, and passes through the first beam splitter to be divided into a first light beam and a second light beam. The first light beam is interfered and then enters the phase detector through the third beam splitter, and the second light beam in the left exposure beam is incident on the holographic grating substrate to form first interference fringes for exposure.

(25) A right exposure beam is reflected by the third reflector and the fourth reflector in sequence, passes through the second phase modulator, and passes through the second beam splitter to be divided into a third light beam and a fourth light beam. The third light beam is interfered and then enters the phase detector through the third beam splitter, and the fourth light beam is incident on the holographic grating substrate to form second interference fringes for exposure. The first interference fringes and the second interference fringes jointly form the interference fringes.

(26) When the holographic grating substrate and the two-dimensional worktop move, the reference mirror and the displacement measurement apparatus position the two-dimensional worktop and detect the displacement of the two-dimensional worktop in real time. The master controller controls the first phase modulator and the second phase modulator to modulate phases of the first interference fringes and phases of the second interference fringes when the two-dimensional worktop performs stepping scanning motion to any point, so that phases of modulated interference fringes meets the phase distribution of the predetermined phases of the interference fringes.

(27) Compared with the prior art, the present disclosure adopts the interference fringes from the scanning beam interference lithography system to modulate phases of the holographic grating substrate, thereby compensating for the surface error of the holographic grating substrate, solving the surface error of the holographic grating substrate due to processing accuracy limitations in the field of holographic grating research and development, and improving full-aperture diffraction wavefront quality of a grating. Through coordination between feedback of the displacement measurement apparatus for the position of the two-dimensional worktop and the modulation of the scanning beam interference lithography system, full-aperture compensation of the surface error of the holographic grating substrate is achieved, which not only reduces processing accuracy requirements for the holographic grating substrate, but also improves the full-aperture diffraction wavefront quality of the grating. Moreover, the present disclosure is conducive to shortening a processing period of a meter-level grating substrate and reducing costs.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a flow chart of a method for compensating surface error of a holographic grating substrate based on scanning and exposure technology according to one embodiment of the present

disclosure.

(2) FIG. 2 is a schematic diagram of the surface error of the method according to one embodiment of the present disclosure.

(3) FIG. 3 is a schematic diagram showing the holographic grating substrate is process by stepping, scanning, and exposure according to one embodiment of the present disclosure.

(4) FIG. 4 is a schematic diagram of a device for compensating surface error of the holographic grating substrate based on the scanning and exposure technology according to one embodiment of the present disclosure.

(5) Reference numbers in the drawings: right exposure beam **1**; left exposure beam **2**; first reflector **3**; second reflector **4**; third reflector **5**; fourth reflector **6**; first phase modulator **7**; second phase modulator **8**; first beam splitter **9**; second beam splitter **10**; holographic grating substrate **11**; two-dimensional worktop **12**; air bearing table **13**; reference mirror **14**; displacement detection apparatus **15**; third beam splitter **16**; phase detector **17**; and master controller **18**.

DETAILED DESCRIPTION

(6) Embodiments of the present disclosure are described with reference to the accompanying drawings as follow. In the following description, same modules are denoted by same reference numerals. In the case of the same reference numerals, their names and functions are the same. Therefore, the detailed description thereof is not repeated.

(7) In order to make objectives, technical solutions, and advantages of the present disclosure clearer, the following further describes the present disclosure in detail with reference to accompanying drawings and embodiments. It should be understood that the specific embodiments described here are only used to explain the present disclosure, but not to limit the present disclosure.

(8) As shown in FIG. 1, the present disclosure provides a method for compensating surface error of a holographic grating substrate **11** based on scanning and exposure technology. The method comprises steps **S101-S103**.

(9) The step **S101** comprises obtaining a surface error distribution diagram corresponding to the holographic grating substrate **11** by detecting the surface error of the holographic grating substrate **11**.

(10) The step **S102** comprises obtaining a phase distribution of predetermined phases compensating for the surface error through simulation calculation.

(11) The step **S103** comprises decomposing the phase distribution of the predetermined phases into a first coordinate matrix corresponding to phases of the holographic grating substrate **11** in a point-to-point manner, exposing coordinate points of the predetermined phases, different with the phases of the holographic grating substrate **11**, in the first coordinate matrix on photoresist disposed on a surface of the holographic grating substrate **11** by stepping, scanning, and exposure, and developing to obtain a grating mask.

(12) In one optional embodiment, the surface error of the holographic grating substrate **11** is detected by an interferometer to obtain a wavefront map (i.e., the surface error distribution diagram) corresponding to the surface error of the holographic grating substrate **11**. The phase distribution of predetermined phases compensating for the surface error is obtained through simulation calculation. Phases of the surface error are pre-compensated by using a scanning interference field exposure model. A compensation table is formed by pre-compensated phase values of the phases of the surface error. The compensation table is input into a phase control apparatus. The surface error is compensated from an exposure start point to an exposure end point. The surface error is compensated point by point according to the compensation table, and finally an overall diffraction wavefront that is close to an ideal diffraction wavefront is obtained.

(13) Specifically, the holographic grating substrate **11** of a predetermined size is accurately positioned from the exposure start point to the exposure end point by a displacement measurement apparatus for a two-dimensional worktop **12** to obtain a second coordinate matrix of the

holographic grating substrate **11**. Then, the predetermined phases of the surface error of the holographic grating substrate **11** is one-to-one corresponding to coordinate points in the second coordinate matrix of the holographic grating substrate **11**. A position of the holographic grating substrate **11** is feedback by the displacement measurement apparatus in real time, and the photoresist on the surface of the holographic grating substrate **11** is scanned, exposed, and developed to obtain the grating mask after surface compensation modulation. It should be noted that the detection of the surface error of the holographic grating substrate **11** is not limited to be performed by the interferometer, and those skilled in the art can flexibly choose other equipment, which is not limited thereto.

(14) The present disclosure adopts the interference fringes from the scanning beam interference lithography system to modulate the phases of the holographic grating substrate **11**, thereby compensating for the surface error of the holographic grating substrate **11**, solving the surface error of the holographic grating substrate **11** due to processing accuracy limitations in the field of holographic grating research and development, and improving full-aperture diffraction wavefront quality of a grating. Through coordination between feedback of the displacement measurement apparatus for the position of the two-dimensional worktop and the modulation of the scanning beam interference lithography system, full-aperture compensation of the surface error of the holographic grating substrate **11** is achieved, which not only reduces processing accuracy requirements for the holographic grating substrate **11**, but also improves the full-aperture diffraction wavefront quality of the grating. Moreover, the present disclosure is conducive to shortening a processing period of a meter-level grating substrate and reducing costs.

(15) In one optional embodiment, after the step of obtaining the phase distribution of the predetermined phases compensating for the surface error through simulation calculation, the method further comprises a step of pre-compensating phases of the surface error by using a scanning interference field exposure model, and forming a compensation table by pre-compensated phase values of the phases of the surface error.

(16) In one optional embodiment, the step of decomposing the phase distribution of the predetermined phases into the first coordinate matrix corresponding to phases of the holographic grating substrate in the point-to-point manner, exposing the coordinate points of the predetermined phases, different with the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on the surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain the grating mask comprises: accurately positioning the holographic grating substrate of a predetermined size from an exposure start point to an exposure end point by a displacement measurement apparatus to obtain a second coordinate matrix of the holographic grating substrate; and corresponding the predetermined phases for compensating the surface error of the holographic grating substrate one-to-one to the coordinate points in the second coordinate matrix of the holographic grating substrate, feeding back a position of the holographic grating substrate by the displacement measurement apparatus in real time, and scanning and exposing the photoresist on the surface of the holographic grating substrate by a scanning beam interference lithography system, and then developing the photoresist to obtain the grating mask after surface compensation modulation.

(17) In one optional embodiment, a phase modulation principle of the scanning beam interference lithography system is as follow.

(18) First, spatially filtered and collimated small-size Gaussian beams emitted by the scanning beam interference lithography system are interfered at beam waist positions thereof to form interference fringes. A phase of the interference fringes at a position (x,y) at time t is $\phi_{\text{sub.ph}}(x,y,t)$, a total exposure after scanning and exposure is:

$$(19) D(x,y) = A(x,y) \{ 1 + \gamma \cos[\frac{2}{P}x + \phi_{\text{ph}}(x,y)] \}.$$

(20) The spatially filtered and collimated small-size Gaussian beams comprises a left exposure beam and a right exposure beam. An energy distribution of the left exposure beam is consistent

with an energy distribution of the right exposure beam.

$$(21) \frac{2}{p} + \varphi_{ph}(x, y)$$

is an exposure phase distribution after stepping, scanning, and exposure, and $\Phi_{sub.ph}(x, y)$ is a predetermined phase distribution related to $\varnothing_{sub.ph}(x, y, t)$, and a value of $\varnothing_{sub.ph}(x, y, t)$ is changed to obtain the predetermined phase distribution $\Phi_{sub.ph}(x, y)$ of the surface of the holographic grating substrate. $A(x, y)$ represents energy of the left exposure beam and the right exposure beam. $A(x, y)$ is expressed as follow:

$$(22) A(x, y) = A_y \sqrt{\frac{2}{\pi}} \exp\left\{-2 \frac{[x - (n-1)S]^2}{\frac{2}{\lambda}}\right\}.$$

(23) Then, detecting the surface error of the holographic grating substrate, and obtaining the phase distribution of the predetermined phases compensating for the surface error through simulation calculation:

$$(24) \varnothing_{ph,e}(x, y) = \begin{bmatrix} \varnothing_{ph,e}(x_1, y_1) & \varnothing_{ph,e}(x_m, y_1) \\ \varnothing_{ph,e}(x_1, y_n) & \varnothing_{ph,e}(x_m, y_n) \end{bmatrix}.$$

(25) Measuring a displacement of a two-dimensional worktop by a displacement measurement apparatus and obtaining the second coordinate matrix H of the holographic grating substrate:

$$(26) H = \begin{bmatrix} x_1, y_1 & x_m, y_1 \\ x_1, y_n & x_m, y_n \end{bmatrix}.$$

(27) During a scanning and stepping process of the two-dimensional worktop, when not modulated, there is a phase distribution W of the holographic grating substrate one-to-one corresponding to coordinate points in the second coordinate matrix H . The phase distribution W is represented by:

$$(28) W = \begin{bmatrix} W(x_1, y_1) & W(x_m, y_1) \\ W(x_1, y_n) & W(x_m, y_n) \end{bmatrix}.$$

(29) φ is a phase modulation term and is a feedback of a phase control apparatus through a displacement measurement of the two-dimensional worktop. Phases of the interference fringes are directionally modulated by changing frequencies of phase modulators. φ is expressed as:

$$(30) \varphi = \begin{bmatrix} \varphi(1, 1) & \varphi(m, 1) \\ \varphi(1, n) & \varphi(m, n) \end{bmatrix}.$$

(31) After phase modulation, a phase distribution $W_{sub.n}$ of modulated phases one-to-one corresponding to the coordinate points in the second coordinate matrix is represented by:

$$(32) W_n = \begin{bmatrix} W_{sub.n}(x_1, y_1) + \varphi(1, 1) & W_{sub.n}(x_m, y_1) + \varphi(m, 1) \\ W_{sub.n}(x_1, y_n) + \varphi(1, n) & W_{sub.n}(x_m, y_n) + \varphi(m, n) \end{bmatrix} = \varnothing_{ph,e}(x, y).$$

(33) The phase distribution $W_{sub.n}$ of the modulated phases is same as the phase distribution $\Phi_{sub.ph,e}(x, y)$ of the predetermined phases. Thereby, purposes of compensating the surface error of the holographic grating substrate **11** and improving the diffraction wavefront quality are achieved through the point-to-point modulation method of scanning interference field exposure.

(34) Specifically, when the surface error of the holographic grating substrate **11** is detected by the interferometer, a surface error distribution diagram as shown in FIG. 2 is obtained The phase distribution of predetermined phases compensating for the surface error is obtained through

simulation calculation. As shown in FIG. 3, the phase distribution of the predetermined phases is decomposed into the first coordinate matrix of $m \times n$ corresponding to the second coordinate matrix of the holographic grating substrate in the point-to-point manner. The scanning beam interference lithography system performs step-scanning exposure, and coordinate points of the the phases of the second coordinate matrix that are different with the coordinate points of the predetermined phases in the first coordinate matrix are modulated to be same as the predetermined phases and are respectively recorded on the photoresist on the surface of the holographic grating substrate **11**, and finally the grating mask is obtained.

(35) The method for compensating the surface error of the holographic grating substrate **11** based on the scanning and exposure technology of the present disclosure is shown to be practical and effective through simulation and experiment. Specifically, the surface error of the holographic grating substrate **11** is $PV=3.244\lambda$, and 85 mm $\times$ 85 mm grating mask exposure experiment of a +1-order diffraction wavefront error $PV=2.573\lambda$ is carried out to measure the surface error. The phase distribution of the interference fringes for compensating the surface error is calculated. During an exposure process, dynamic phase locking technology is adopted to reflect the predetermined phases in the diffraction wavefront of the grating mask. An actual diffraction wavefront of the grating mask is measured to be $PV=0.399\lambda$, which greatly reduces a PV value of the surface error of the holographic grating substrate **11**.

(36) The method for compensating the surface error of the holographic grating substrate **11** based on the scanning and exposure technology of the present disclosure is used to solve a problem of a surface processing error of the holographic grating substrate **11** that cannot be eliminated. In the scanning, splicing, and exposure process, an error caused by the surface error of the holographic grating substrate **11** to the diffraction wavefront is compensated by the phase modulation of the interference fringes. In the field of holographic gratings, the method based on the scanning and exposure technology effectively improves the diffraction wavefront quality of the holographic grating substrate **11**, such as a full-aperture grating substrate **11**, while reducing processing accuracy requirements of the holographic gratings of various sizes, improving processing efficiency of the holographic grating substrate **11**, and shortening a circle of grating development.

(37) As shown in FIG. 4, the present disclosure provides a device for compensating the surface shape error of the holographic grating substrate **11** based on the scanning and exposure technology. The device has scan and exposure functions and is applied to the method mentioned above. The device comprises a first reflector **3**, a second reflector **4**, a third reflector **5**, a fourth reflector **6**, a first phase modulator **7**, a second phase modulator **8**, a first beam splitter **9**, a second beam splitter **10**, a third beam splitter **16**, a two-dimensional worktop **12**, a reference mirror **14**, a displacement detection apparatus **15**, a phase detector **17**, and a master controller **18**. The two-dimensional worktop **12** is disposed on an air bearing table **13**, which further reduce an influence of vibration generated during operation on positioning of the displacement detection apparatus **15**.

(38) A left exposure beam **2** is reflected by the first reflector **3** and the fourth reflector **6** in sequence, passes through the first phase modulator **7**, and passes through the first beam splitter **9** to be divided into a first light beam and a second light beam. The first light beam is interfered and then enters the phase detector **17** through the third beam splitter **16**. The phase detector **17** performs real-time phase monitoring and dynamic phase locking on the first light beam by heterodyne measurement or homodyne measurement. The second light beam is incident on the holographic grating substrate **11** to form first interference fringes for exposure.

(39) A right exposure beam **1** is reflected by the third reflector **5** and the fourth reflector **6** in sequence, passes through the second phase modulator **8**, and passes through the second beam splitter **10** to be divided into a third light beam and a fourth light beam. The third light beam is interfered and then enters the phase detector **17** through the third beam splitter **16**. The phase detector **17** performs real-time phase monitoring and dynamic phase locking on the third light beam by heterodyne measurement or homodyne measurement. The fourth light beam is incident on the

holographic grating substrate **11** to form second interference fringes for exposure.

(40) When the holographic grating substrate **11** and the two-dimensional worktop **12** move, the reference mirror **14** and the displacement measurement apparatus **15** position the two-dimensional worktop **12** and detect the displacement of the two-dimensional worktop **12** in real time. The master controller **18** controls the first phase modulator **7** to modulate phases of the first interference fringe and the second phase modulator **8** to modulate phases of the second interference fringes, when the two-dimensional worktop **12** performs stepping scanning motion to any coordinate point, so that phases of modulated interference fringes on the holographic grating substrate **11** meets the phase distribution of the predetermined phases of the interference fringes.

(41) In the device for compensating the surface shape error of the holographic grating substrate **11** based on the scanning and exposure technology of the present disclosure, the holographic grating substrate **11** of the predetermined size is accurately positioned from the exposure start point to the exposure end point by the displacement measurement apparatus **15** for the two-dimensional worktop **12** to obtain the second coordinate matrix of the holographic grating substrate **11**. Then, the predetermined phases for compensating the surface error are one-to-one corresponding to the coordinate points in the second coordinate matrix of the holographic grating substrate **11**, the position of the holographic grating substrate **11** is feedback by the displacement measurement apparatus **15** in real time. After scanning and exposing the photoresist on the surface of the holographic grating substrate **11**, the grating mask after surface compensation modulation is obtained, solving the surface error of the holographic grating substrate **11** due to processing accuracy limitations in the field of holographic grating research and development, and improving full-aperture diffraction wavefront quality of the grating. Through coordination between feedback of the displacement measurement apparatus for the position of the two-dimensional worktop and the modulation of the scanning beam interference lithography system, full-aperture compensation of the surface error of the holographic grating substrate **11** is achieved, which not only reduces processing accuracy requirements for the holographic grating substrate **11**, but also improves the full-aperture diffraction wavefront quality of the grating.

(42) It should be understood that various forms of the processes shown above may be reorder and the steps may be added or removed. For example, the steps described in the present disclosure may be performed at the same time, sequentially, or in a different order, as long as a desired result of technical solutions disclosed in the present disclosure can be implemented, which is not limited herein.

(43) The above specific embodiments do not constitute a limitation on the protection scope of the present disclosure. Those skilled in the art should understand that various modifications, combinations, sub-combinations and substitutions may be made according to design requirements and other factors. Any modification, equivalent replacement and improvement made within the spirit and principle of the present disclosure shall be included in the protection scope of the present disclosure.

Claims

1. A method for compensating a surface error of a holographic grating substrate, comprising steps: obtaining a surface error distribution diagram corresponding to the holographic grating substrate by detecting the surface error of the holographic grating substrate; obtaining a phase distribution of predetermined phases compensating for the surface error through simulation calculation; and decomposing the phase distribution of the predetermined phases into a first coordinate matrix corresponding to phases of the holographic grating substrate in a point-to-point manner, exposing coordinate points of the predetermined phases, different from the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on a surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain a grating mask;

wherein a phase modulation principle of a scanning beam interference lithography system comprises: interfering at beam waist positions of spatially filtered and collimated small-size Gaussian beams emitted by the scanning beam interference lithography system to form interference fringes; a phase of the interference fringes at a position (x,y) at time t is $\phi.\text{sub.ph}(x,y,t)$, a total exposure after scanning and exposure is:

$D(x,y)=A(x,y)\{1+\gamma \cos[2\pi/p_x+\Phi.\text{sub.ph}(x,y)]\}$, wherein $2\pi/p_x+\Phi.\text{sub.ph}(x,y)$ is an exposure phase distribution after stepping, scanning, and exposure, and $\Phi.\text{sub.ph}(x,y)$ is a phase distribution related to $\phi.\text{sub.ph}(x,y,t)$, and a value of $\phi.\text{sub.ph}(x,y,t)$ is changed to obtain a predetermined phase distribution $\Phi.\text{sub.ph}(x,y)$ of the surface of the holographic grating substrate; detecting the surface error of the holographic grating substrate, and obtaining the phase distribution $\Phi.\text{sub.ph,e}(x,y)$ of the predetermined phases compensating for the surface error through the

simulation calculation:
$$\text{ph},e(x,y) = \begin{bmatrix} \text{Math.} & \text{Math.} & \text{Math.} \\ (x_1,y_1) & (x_m,y_1) & \\ & \text{Math.} & \\ & (x_1,y_n) & (x_m,y_n) \end{bmatrix}$$
; measuring a displacement of

a two-dimensional worktop by a displacement measurement apparatus and obtaining a second

coordinate matrix H of the holographic grating substrate:
$$H = \begin{bmatrix} \text{Math.} & \text{Math.} & \text{Math.} \\ x_1,y_1 & x_m,y_1 & \\ & \text{Math.} & \\ & x_1,y_n & x_m,y_n \end{bmatrix}$$
;

wherein during a scanning and stepping process of the two-dimensional worktop, when not modulated, there is a phase distribution W of the phases of the holographic grating substrate one-to-one corresponding to coordinate points in the second coordinate matrix H, the phase distribution

W is represented by:
$$W = \begin{bmatrix} \text{Math.} & \text{Math.} & \text{Math.} \\ (x_1,y_1) & (x_m,y_1) & \\ & \text{Math.} & \\ & (x_1,y_n) & (x_m,y_n) \end{bmatrix}$$
; ϕ is a phase modulation term and is a

feedback of a phase control apparatus through a displacement measurement of the two-dimensional worktop; phases of the interference fringes are directionally modulated by changing frequencies of

phase modulators, ϕ is expressed as:
$$= \begin{bmatrix} \text{Math.} & \text{Math.} & \text{Math.} \\ (1,1) & (m,1) & \\ & \text{Math.} & \\ & (1,n) & (m,n) \end{bmatrix}$$
; after phase modulation, a

phase distribution W.sub.n of modulated phases one-to-one corresponding to the coordinate points in the second coordinate matrix is represented by:

$$W_n = \begin{bmatrix} \text{Math.} & \text{Math.} & \text{Math.} \\ (x_1,y_1) + (1,1) & (x_m,y_1) + (m,1) & \\ & \text{Math.} & \\ & (x_1,y_n) + (1,n) & (x_m,y_n) + (m,n) \end{bmatrix} = \text{ph},e(x,y)$$
, wherein the phase

distribution W.sub.n of the modulated phases is same as the phase distribution $\Phi.\text{sub.ph,e}(x,y)$ of the predetermined phases; the step of decomposing the phase distribution of the predetermined phases into the first coordinate matrix corresponding to phases of the holographic grating substrate in the point-to-point manner, exposing the coordinate points of the predetermined phases, different from the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on the surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain the grating mask comprises: accurately positioning the holographic grating substrate of a predetermined size from an exposure start point to an exposure end point by the displacement measurement apparatus to obtain the second coordinate matrix of the holographic

grating substrate; and corresponding the predetermined phases for compensating the surface error of the holographic grating substrate one-to-one to the coordinate points in the second coordinate matrix of the holographic grating substrate, feeding back a position of the holographic grating substrate by the displacement measurement apparatus in real time, and scanning and exposing the photoresist on the surface of the holographic grating substrate by a scanning beam interference lithography system, and then developing the photoresist to obtain the grating mask after surface compensation modulation.

2. The method according to claim 1, wherein after the step of obtaining the phase distribution of the predetermined phases compensating for the surface error through the simulation calculation, the method further comprises a step: pre-compensating phases of the surface error by using a scanning interference field exposure model, and forming a compensation table by pre-compensated phase values of the phases of the surface error.

3. A device for compensating a surface error of a holographic grating substrate having scan and exposure functions, comprising: a first reflector, a second reflector, a third reflector, a fourth reflector, a first phase modulator, a second phase modulator, a first beam splitter, a second beam splitter, a third beam splitter, the two-dimensional worktop, a reference mirror, a displacement detection apparatus, a phase detector, and a master controller; wherein the device is configured to execute steps: obtaining a surface error distribution diagram corresponding to the holographic grating substrate by detecting the surface error of the holographic grating substrate; obtaining a phase distribution of predetermined phases compensating for the surface error through simulation calculation; and decomposing the phase distribution of the predetermined phases into a first coordinate matrix corresponding to phases of the holographic grating substrate in a point-to-point manner, exposing coordinate points of the predetermined phases, different from the phases of the holographic grating substrate, in the first coordinate matrix on photoresist disposed on a surface of the holographic grating substrate by stepping, scanning, and exposure, and developing to obtain a grating mask; wherein a phase modulation principle of a scanning beam interference lithography system comprises: interfering at beam waist positions of spatially filtered and collimated small-size Gaussian beams emitted by the scanning beam interference lithography system to form interference fringes; a phase of the interference fringes at a position (x,y) at time t is $\phi_{\text{sub.ph}}(x,y,t)$, a total exposure after scanning and exposure is:

$D(x,y)=A(x,y)\{1+\gamma \cos[2\pi/p_x+\Phi_{\text{sub.ph}}(x,y)]\}$, wherein $2\pi/p_x+\Phi_{\text{sub.ph}}(x,y,t)$ is an exposure phase distribution after stepping, scanning, and exposure, and $\Phi_{\text{sub.ph}}(x,y)$ is a phase distribution related to $\phi_{\text{sub.ph}}(x,y,t)$, and a value of $\phi_{\text{sub.ph}}(x,y,t)$ is changed to obtain a predetermined phase distribution $\Phi_{\text{sub.ph}}(x,y)$ of the surface of the holographic grating substrate; detecting the surface error of the holographic grating substrate, and obtaining the phase distribution $\Phi_{\text{sub.ph,e}}(x,y)$ of the predetermined phases compensating for the surface error through the simulation

calculation:
$$\phi_{\text{ph,e}}(x,y) = \begin{bmatrix} \text{.Math.} & \text{.Math.} & \text{.Math.} \\ (x_1,y_1) & (x_m,y_1) & (x_1,y_n) \end{bmatrix}$$
; measuring a displacement of a two-dimensional worktop by a displacement measurement apparatus and obtaining a second coordinate

matrix H of the holographic grating substrate:
$$H = \begin{bmatrix} \text{.Math.} & \text{.Math.} & \text{.Math.} \\ (x_1,y_1) & (x_m,y_1) & (x_1,y_n) \end{bmatrix}$$
; wherein

during a scanning and stepping process of the two-dimensional worktop, when not modulated, there is a phase distribution W of the phases of the holographic grating substrate one-to-one corresponding to coordinate points in the second coordinate matrix H, the phase distribution W is

represented by: $W = [\begin{matrix} \text{.Math.} & \text{.Math.} & \text{.Math.} \\ (x_1, y_1) & (x_m, y_1) & (x_1, y_n) & (x_m, y_n) \end{matrix}]$; φ is a phase modulation term and is a

feedback of a phase control apparatus through a displacement measurement of the two-dimensional worktop; phases of the interference fringes are directionally modulated by changing frequencies of

phase modulators, φ is expressed as: $\varphi = [\begin{matrix} \text{.Math.} & \text{.Math.} & \text{.Math.} \\ (1, 1) & (m, 1) & (1, n) & (m, n) \end{matrix}]$; after phase modulation, a

phase distribution $W_{\text{sub}.n}$ of modulated phases one-to-one corresponding to the coordinate points in the second coordinate matrix is represented by:

$$W_n = [\begin{matrix} \text{.Math.} & \text{.Math.} & \text{.Math.} \\ (x_1, y_1) + (1, 1) & (x_m, y_1) + (m, 1) & (x_1, y_n) + (1, n) & (x_m, y_n) + (m, n) \end{matrix}] = \varphi_{\text{ph}, e}(x, y), \text{ wherein the phase}$$

distribution $W_{\text{sub}.n}$ of the modulated phases is same as the phase distribution $\Phi_{\text{sub}.ph, e}(x, y)$ of the predetermined phases: wherein a left exposure beam is reflected by the first reflector and the fourth reflector in sequence, passes through the first phase modulator, and passes through the first beam splitter to be divided into a first light beam and a second light beam; the first light beam is interfered and then enters the phase detector through the third beam splitter, and the second light beam in the left exposure beam is incident on the holographic grating substrate to form first interference fringes for exposure; wherein a right exposure beam is reflected by the third reflector and the fourth reflector in sequence, passes through the second phase modulator, and passes through the second beam splitter to be divided into a third light beam and a fourth light beam; the third light beam is interfered and then enters the phase detector through the third beam splitter, and the fourth light beam is incident on the holographic grating substrate to form second interference fringes for exposure; when the holographic grating substrate and the two-dimensional worktop move, the reference mirror and the displacement measurement apparatus position the two-dimensional worktop and detect the displacement of the two-dimensional worktop in real time; wherein the master controller controls the first phase modulator and the second phase modulator to modulate phases of the first interference fringes and phases of the second interference fringes when the two-dimensional worktop performs stepping scanning motion to any point, so that phases of the modulated interference fringes meet the phase distribution of the predetermined phases of the interference fringes.
